

Login with username and password

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1. Frontend

1.1 Overview

The login feature provides a secure and user-friendly way for users to access their accounts. It supports standard email-based authentication with built-in validation to ensure data quality, as well as a convenient "Sign in with Google" option for faster access. The interface is designed to be responsive, providing immediate feedback if information is entered incorrectly or if the system is currently processing a request.

1.2 Step-by-Step Flow

1. Data Entry: The user enters their email and password into the respective fields. The system automatically cleans up any accidental spaces at the beginning or end of the input.
2. Validation: As the user types, the system checks that the email follows a standard format and that the password meets security requirements (containing uppercase, lowercase, numbers, and special characters).
3. Visibility Control: The user can click a button to toggle the password visibility between hidden (masked) and plain text to ensure accuracy while typing.
4. Submission: Once both fields are valid, the "Login" button becomes active. When clicked, the system triggers the authentication process.
5. Processing: The button displays a "Loading..." state to prevent the user from accidentally submitting the form multiple times.
6. Outcome:

* If the credentials are correct, the user is redirected to the dashboard.

* If the credentials are incorrect or an error occurs, the system displays a generic error message to maintain security.

7. Alternative Login: Alternatively, the user can click "Sign in with Google," which redirects them to the Google authentication page to complete the login process.

1.3 Key Concepts

- Input Masking: A security feature that hides the password characters (usually as dots or asterisks) while the user types to prevent shoulder-surfing.
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Form Validation: Automated checks that ensure the data entered by the user meets the required format before the system attempts to process it.

- Loading State: A visual indicator that tells the user the system is working on their request, which also serves to prevent duplicate submissions.
- Data Sanitization: The automatic removal of unnecessary spaces from inputs to prevent common entry errors.
- Generic Error Handling: A security practice where the system provides a vague error message (e.g., "Invalid email or password") rather than specifying which field was wrong, preventing attackers from guessing valid account details.

1.4 Business Value

- Enhanced Security: By enforcing strict password complexity and masking inputs, we protect user accounts from unauthorized access.
- Improved User Experience: Features like the password visibility toggle and real-time validation reduce user frustration by helping them identify and fix errors immediately.
- Reduced Support Overhead: By preventing invalid data from being submitted and providing clear, immediate feedback, we reduce the number of login-related support tickets.
- Increased Conversion: Offering "Sign in with Google" lowers the barrier to entry, making it faster and easier for users to access the platform.

2. Logic

2.1 Overview

This feature enables a seamless "Sign in with Google" experience for our users. It automates the account creation and login process by securely verifying a user's identity through Google and syncing their profile information with our internal system. This ensures that users can access our platform quickly without needing to manage a separate password for our application.

2.2 Step-by-Step Flow

1. The user initiates the login process by selecting the Google sign-in option, which redirects them to Google to verify their identity.
2. Once verified, Google sends the user's profile information (such as their name and email address) back to our system.
3. Our system checks our database to see if the user already has an account associated with that email address.
4. If the user is new, we automatically create a new account for them using the information provided by Google.

5. If the user already exists, we confirm their account details and update their login method to reflect that they are using Google authentication.
6. Finally, the system generates a secure session token, allowing the user to remain logged in and access the platform for the next 24 hours.

2.3 Key Concepts

- Google OAuth: A secure industry-standard protocol that allows users to sign into our application using their existing Google credentials without sharing their Google password with us.
- User Synchronization: The process of matching or creating a user record in our database based on the information received from Google, ensuring the user's identity is consistent.
- Session Token: A temporary, secure digital "key" issued to the user after a successful login, which allows them to navigate our site without having to log in repeatedly.

2.4 Business Value

- Improved User Experience: Reduces friction during onboarding by allowing users to sign up or log in with a single click.
- Increased Conversion: Lowering the barrier to entry encourages more visitors to register and engage with our platform.
- Enhanced Security: By leveraging Google's robust authentication infrastructure, we reduce the risks associated with managing user passwords and decrease the likelihood of account-related security issues.

3. Data

3.1 Overview

This feature manages the user login process, ensuring that only authorized users can access the system while protecting accounts from unauthorized access attempts. It includes a security mechanism that temporarily restricts access to an account if multiple incorrect password attempts are made.

3.2 Step-by-Step Flow

1. Request Submission: The user submits their username and password through the login interface.
2. Account Verification: The system checks if the username exists in the database. If the account is already locked due to previous failed attempts, the login is rejected immediately.
3. Credential Validation: The system compares the provided password against the stored record.
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Handling Failures: If the password is incorrect, the system records the failed attempt. If the total number of failed attempts reaches five, the account is automatically locked for security.

5. Successful Login: If the credentials are correct, the system clears any previous failed attempt counts and generates a secure authentication token, which allows the user to access the system.

3.3 Key Concepts

- Authentication Token: A secure digital key provided to the user after a successful login, which acts as proof of identity for subsequent actions.
- Account Locking: A security measure that prevents further login attempts on an account after five consecutive failed tries to protect against unauthorized access.
- Credential Validation: The process of verifying that the username and password provided match the information stored in our secure database.
- Failed Attempt Tracking: A background counter that monitors incorrect login attempts to trigger security protocols when necessary.

3.4 Business Value

- Enhanced Security: By automatically locking accounts after repeated failed attempts, we significantly reduce the risk of "brute-force" attacks where unauthorized parties try to guess passwords.
- Improved User Experience: The system provides clear feedback when a login fails, ensuring users know why they cannot access their account.
- Operational Efficiency: The automated nature of the locking and resetting process reduces the need for manual administrative intervention, allowing users to regain access or manage their accounts seamlessly once they provide the correct credentials.